Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/01/2021
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Round Robin - Thermophysical Property Measurement		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	04/01/2017	**End Date:**	03/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Krivilev, Mikhail Ph.D. (Co-PI: Udmurt State University Izhev, Russia)		
Grant/Contract No.:	NNX17AH41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Researchers go to space to conduct material property measurements that are required to model terrestrial manufacturing processes to provide higher performance and discover new materials to make people's lives better around the world. We justify the expense of doing tests in microgravity by concentrating on those materials which are impossible to evaluate on Earth or have properties that can only be measured with higher precision when gravitational accelerations are minimized. We know we get better results but we do not yet know how much better. The JAXA (Japanese Aerospace Exploration Agency) Electrostatic Levitation Furnace (ELF) was designed to process high temperature oxides but it can be readily adapted to allow testing of industrially important metals and alloys as well. By selecting materials that are already being processed using another technique, the European Space Agency (ESA) Electromagnetic Levitation facility on the International Space Station (ISS), these two facilities can be compared so that researchers know their relative strengths such that they may complement each other to maximize science output from space tests. Our objectives are to conduct space experiments to compare to ground results on a baseline material set and to model the influence of gravity on measurement techniques by various research groups across the world. Our approach is to employ Round Robin testing by multiple researchers on the same material using different techniques, on different facilities, with different analysis protocols to compare results. The significance of this work is that we will then know the relative increase in measurement fidelity and which material works best in which facility to efficiently use the limited available space sciences resources such as launch cost, crew time, and telemetry bandwidth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program leverages the existing ESA International Topical Team organization to develop a strategy to compare the performance of current flight and ground facilities investigating thermophysical property measurement using levitation techniques. New experiments are proposed for the JAXA-ELF facility on the ISS to complement other ongoing investigations that use both the ISS-based Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) and ground-based facilities. We will efficiently use available resources, both fiscal and crew time, by selecting materials compatible with those previously prioritized by other existing research groups. In doing so, we will expand the capabilities of space electrostatic levitation processing to allow for investigation of additional thermophysical material properties and manifesting new material classes across microgravity platforms. The key scientific questions we address are: (1) How do we control and minimize gravity-induced systematic error in thermophysical property measurement and does theory explain the observed deviations in results? (2) For which material classes are space-based platforms delivering thermophysical property measurement results with higher accuracy and/or precision than ground-based facilities? Thermophysical property testing in microgravity provides enhanced fidelity of results purportedly due to three main effects. First, without strong gravitational accelerations and with reduced levitation forces a more spherical sample allows for better analysis of experimental behavior. Second, better control of convection in space results in higher measurement precision. Third, sedimentation and buoyancy induced segregation are eliminated in reduced gravity. But how much better do we do in space as compared to ground? The obvious solution is to quantify variability across platforms – both in space and on ground.
Task Progress:	Highlights of Work Accomplished During Grant Period Mar 2020- Feb 2021. • Published eight peer-reviewed articles (see Bibliography section). • Graduated two MS students (Azersky and Dytiuk). • Sent spare JAXA flight samples to LUVAK test laboratory to analyze and document sample composition and oxygen contamination levels. • Prepared and delivered samples for parallel ground-based testing at Tohoku University in Sendai Japan, DLR (German Aerospace Center) in Köln Germany, and TU-Graz (Technische Universität Graz) in Graz, Austria. • Attended TMS (The Minerals, Metals & Materials Society) 149th Annual Meeting, 23-27 February, 2020, in San Diego ,CA, as Materials Science in Reduced Gravity symposium organizer and presenter. • Attended ASGSR-36 (American Society for Gravitational and Space Research) 2020 Annual Conference, November 4-9, 2020, virtual conference as national president, symposium organizer, and presenter. • Featured speaker at Japan Microgravity Application Society Conference JASMAC-32, and 181st Space Studies Board Meeting. • Conducted flight testing during ISS expedition 63 and 64 using JAXA ELF facility ; Successfully finished Sample holder S/N 006 and S/N 007, initiating S/N 008. Highlights of Previous Work Accomplished. • Finalized preparation of Science Requirements Document (SRD) as the first NASA-Principal Investigator to conduct flight experiments using the JAXA-ELF on the ISS • Fabricated and delivered JAXA flight samples (and pares) to JAXA with the help of the ESL (electrostatic levitation) staff led by Michael SanSoucie of NASA-Marshall Space Flight Center (MSFC).
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Xiao X, Rodriguez JE, Kelton KF, Matson DM, "Statistical learning for evaluation of crystal growth in low-melting alloy droplets with application to quasicrystal-forming Ti-Zr-Ni alloys." Modelling and Simulation in Materials Science and Engineering. 2020 Oct;28(8):085008, 1-22. https://doi.org/10.1088/1361-651X/abbfba , Oct-2020
Articles in Peer-reviewed Journals	Matson DM. "Influence of induced convection on transformation kinetics during rapid solidification of steel alloys: the retained damage model." JOM. 2020 Nov;72(11):4109-16. https://doi.org/10.1007/s11837-020-04375-2 , Nov-2020

Articles in Peer-reviewed Journals	Nawer J, Gosse S, Matson DM. "Tracking evaporation during levitation processing of nickel-based superalloys on the ISS." JOM. 2020 Sep;72(9):3132-9. https://doi.org/10.1007/s11837-020-04256-8 , Sep-2020
Articles in Peer-reviewed Journals	Matson DM. "Maximizing scientific return from shared facilities working in space: ESA's investigator working group for the ISS-EML facility." Int J Microgravity Sci Appl. 2020;37(3):370301, 1-4. https://www.jstage.jst.go.jp/article/ijmsa/37/3/37_370301/_pdf , Jul-2020
Articles in Peer-reviewed Journals	Baker EB, Nawer J, Xiao X, Matson DM. "MHD surrogate model for control of convection in electromagnetically levitated molten metal droplets processed using the ISS-EML facility" NPJ Microgravity. 2020 Mar 16;6:9. https://doi.org/10.1038/s41526-020-0099-7 ; PMID: 32195320; PMCID: PMC7076000 , Mar-2020
Articles in Peer-reviewed Journals	Jeon S, Sansoucie MP, Shuleshova O, Kaban I, and Matson DM. "Density, excess volume and structure of Fe-Cr-Ni melts." J Chem Phys. 2020 Mar;152(9):094501, 1-7. https://doi.org/10.1063/1.5140787 , Mar-2020
Articles in Peer-reviewed Journals	Nawer J, Xiao X, Sansoucie MP, Matson DM. "Effect of mass evaporation on measurement of liquid density of Ni-based superalloys using ground and space levitation techniques." High Temperature High Pressures. 2020;49(1-2):17-29. https://doi.org/10.32908/hthp.v49.839 , Jan-2020
Articles in Peer-reviewed Journals	Bracker GP, Baker EB, Nawer J, Sellers ME, Gangopadhyay AK, Kelton KF, Xiao X, Lee J, Reinartz M, Burgraf S, Herlach D, Rettenmayr M, Matson DM, Hyers RW. "The effect of flow regime on surface oscillations during electromagnetic levitation experiments." High Temperature High Pressures. 2020;49(1-2):49-60. https://doi.org/10.32908/hthp.v49.817 , Jan-2020